

Water used during real art vs ai art (average, done using chatgpt, hope that's ok since this is an defendingaiart sub)

Posted by u/Wof-World • • 0 points • 16 comments

Certainly! Here's the full comparison :

Question:

What uses more water — real art or AI-generated art?

Summary:

Real (traditional + digital) art currently uses more water overall than AI art, primarily due to the water embedded in food consumed by the artist during the time they spend creating. However, AI art may catch up or surpass in total water usage due to how fast and frequently it scales.

Breakdown:

Real Art (Traditional + Digital)

Components considered:

1. Traditional materials (paper, paint, brush cleaning)
2. Power used by digital art software
3. Food consumed by the artist while working

Water usage estimates (annually):

Traditional materials: ~417 billion liters/year

Digital software electricity: ~8 billion liters/year

Food consumed while drawing (~30 billion hours globally):
 $\approx 146 \text{ liters/hour} \times 30 \text{ billion hours} = \sim 4.38 \text{ trillion liters/year}$

→ Total = ~4.805 trillion liters/year

AI Art

Components considered:

1. Power used for AI image generation

2. Food consumed while prompting

Water usage estimates (annually):

AI image generation (~20 billion images): $2 \text{ liters/image} \times 20\text{B} = \sim 40 \text{ billion liters/year}$

Food consumed while prompting (5 minutes per image on average):
 $\approx 1.67 \text{ billion hours} \times 146 \text{ liters/hour} = \sim 244 \text{ billion liters/year}$

→ Total = ~284 billion liters/year

Per 1-Hour Session Comparison

Real Art (1 hour):

Traditional tools (e.g., 2 pages + cleaning): ~21 liters

Power use (digital art): ~0.27 liters

Food consumed: ~146 liters

→ Total: ~167.3 liters

AI Art (1 image taking ~5 minutes):

AI image generation: ~2 liters

Food consumed during 5 minutes: ~12.2 liters

→ Total: ~14.2 liters

Conclusion:

Real art currently uses significantly more water per year (~4.805 trillion liters) than AI art (~284 billion liters).

The main reason is the amount of time human artists spend creating, which increases the amount of food they consume (and the virtual water footprint tied to that food).

On a per-hour or per-session basis, real art uses over 10 times more water than AI art.

However, AI art can be generated rapidly and at scale, so if usage continues to grow, its total environmental impact — including water use — could surpass that of traditional art in the near future.

Let me know if you'd like the same comparison done using carbon emissions, energy consumption, or if you'd like a version using vegan diets or different user behavior scenarios.

Comments

u/Far_Market9582 • 1 points • 2025-06-30 10:34 UTC

I'm all for defending ai and shutting down the bs about ai art being environmentally unsustainable, but these calculations seem really questionable

u/Wof-World • 1 points • 2025-06-30 10:45 UTC

They probably are, I checked them and they aren't that bad tho

u/fullof-salt • 1 points • 2025-06-29 13:45 UTC

What about the water used in making the ai servers and maintenance. Feels a bit weird to exclude it . Also the math is really fucking shaky

u/Vallen_H • 1 points • 2025-06-29 14:38 UTC

But we also gotta include the blood (water-based) that flows in the veins of the human artists, it is not refreshed! It disappears forever...

u/fullof-salt • -1 points • 2025-06-30 08:33 UTC

That feels disingenuous to the actual query like ik you guys can't really think by yourself but not including the huge server costs is strange

u/Far_Market9582 • 1 points • 2025-06-30 11:00 UTC

I think that was satire

u/[deleted] • 1 points • 2025-06-30 09:18 UTC

[removed]

u/Wof-World • 2 points • 2025-06-29 13:56 UTC

I'm pretty sure that's added into the total ai image thing, I'll ask chatgpt

u/fullof-salt • 1 points • 2025-06-30 08:32 UTC

Chatgpt sucks at math can't you like, write down by yourself it's really not that hard

u/Wof-World • 1 points • 2025-06-30 10:44 UTC

Broo 🤔 t, antis say, ai sucks at art can't you like draw it by yourself, it's really not that hard

u/fullof-salt • 2 points • 2025-06-30 17:01 UTC

Yeah but chatgpt s math problems have quite literally been observed and like reported by its own developers tf are you on.

u/Wof-World • 1 points • 2025-06-30 17:52 UTC

I'll check....

u/fullof-salt • 2 points • 2025-06-30 18:11 UTC

That's the spirit , ai can be used as a tool but you also have to do complimentary research to be as accurate as possible when making statements

u/Wof-World • 1 points • 2025-06-30 18:23 UTC

Yesss, thanks ig I'll double check, when I am free again I'll make sure what he/she or whatever ai is said yk

Its
the
best
thing
to
do

u/camelovaty • 7 points • 2025-06-29 12:54 UTC

Oh, so uncomfortable fact for them

u/Wof-World • 3 points • 2025-06-29 12:09 UTC

Note: even without food, and digital art electricity traditional art still takes more water then ai, and with electricity from digital it takes much more time

My research aswell